

Erdős Problem #522

EP-522

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $f(z) = \sum_{0 \leq k \leq n} \epsilon_k z^k$ be a random polynomial, where $\epsilon_k \in \{-1, 1\}$ independently uniformly at random for $0 \leq k \leq n$.

Is it true that, if R_n is the number of roots of $f(z)$ in $\{z \in \mathbb{C} : |z| \leq 1\}$, then $\frac{R_n}{n/2} \rightarrow 1$ almost surely?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>